

Solution

Problem Statement

Let X be the normalized overheating index. The Cumulative Distribution Function (CDF) is:

$$F(x) = \begin{cases} 0, & x \leq 0 \\ kx^2, & 0 < x \leq 1 \\ k(2x - 1), & 1 < x \leq 3 \\ 1, & x > 3 \end{cases}$$

Part (a): Find the constant k

Method 1: CDF Boundary Condition (Standard)

A CDF must satisfy $\lim_{x \rightarrow \infty} F(x) = 1$. Since the support ends at $x = 3$, we must have $F(3) = 1$.

$$\begin{aligned} F(3) &= k(2(3) - 1) \\ 1 &= k(5) \\ k &= 0.2 \end{aligned}$$

Method 2: PDF Normalization (Calculus Approach)

The total area under the Probability Density Function (PDF) must be 1.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Step 1: Derive $f(x)$ symbolically (differentiating $F(x)$).

- $0 < x \leq 1$: $\frac{d}{dx}(kx^2) = 2kx$
- $1 < x \leq 3$: $\frac{d}{dx}(2kx - k) = 2k$

Step 2: Integrate over the domain.

$$\begin{aligned} \int_0^1 2kx \, dx + \int_1^3 2k \, dx &= 1 \\ [kx^2]_0^1 + [2kx]_1^3 &= 1 \\ (k - 0) + (6k - 2k) &= 1 \implies 5k = 1 \implies k = 0.2 \end{aligned}$$

Answer (a)

The constant is $k = 0.2$.

Part (b): Find $P(1 \leq X \leq 2)$ **Method 1: Using the CDF (Fundamental Theorem)**

$$P(a \leq X \leq b) = F(b) - F(a)$$

$$F(2) = 0.2(2(2) - 1) = 0.6$$

$$F(1) = 0.2(1)^2 = 0.2$$

$$P = 0.6 - 0.2 = 0.4$$

Method 2: Integrating the PDF (Direct Area)

Using the density found in Part (c): $f(x) = 0.4$ for $1 < x \leq 3$.

$$\begin{aligned} P(1 \leq X \leq 2) &= \int_1^2 f(x) dx = \int_1^2 0.4 dx \\ &= [0.4x]_1^2 = 0.4(2) - 0.4(1) = 0.4 \end{aligned}$$

Method 3: Complementary Integration

$$P(E) = 1 - P(E^c)$$

We subtract the probability of regions *outside* the interval $[1, 2]$.

$$P(X < 1) = \int_0^1 0.4x dx = [0.2x^2]_0^1 = 0.2$$

$$P(X > 2) = \int_2^3 0.4 dx = [0.4x]_2^3 = 0.4(1) = 0.4$$

$$P(1 \leq X \leq 2) = 1 - (0.2 + 0.4) = 0.4$$

Answer (b)

The probability is **0.4**.

Part (c): Compute the PDF $f(x)$ **Method: Piece-wise Differentiation**

By definition, $f(x) = F'(x)$.

Range 1 ($0 < x \leq 1$):

$$F(x) = 0.2x^2 \implies f(x) = \frac{d}{dx}(0.2x^2) = 0.4x$$

Range 2 ($1 < x \leq 3$):

$$F(x) = 0.2(2x - 1) = 0.4x - 0.2 \implies f(x) = \frac{d}{dx}(0.4x - 0.2) = 0.4$$

Answer (c)

$$f(x) = \begin{cases} 0.4x, & 0 < x \leq 1 \\ 0.4, & 1 < x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$